

How Work Happens, and Where AI Fits In

Session 1



Agenda

- I Introductions, and what this course expects
- II What we will learn today
- III How work happens — the five stages
- IV Mapping our use case
- V Where AI fits, and where it does not
- VI Designing it — Data, Model, Metric, Human
- VII All three steps on one decision, and key takeaways
- VIII Wrap-up

Before we start — let's get to know each other

- 1 Your name**
And what you would like to be called.
- 2 What field are you coming from?**
Your industry and role if you are working — or your program, if you are studying.
- 3 Which best describes you?**
Full-time student · Student who also works · Working professional · Career changer · Something else
- 4 What do you want to walk out of this course able to do?**
One sentence. It shapes what I emphasize over the next fifteen weeks.
- 5 Any prior course with statistics, data analysis, or programming?**
Comfortable in it · Difficult, or a long time ago · None yet — a perfectly normal starting point

Why I am asking

The weekly examples get built around the industries in this room, and your last two answers decide how far we take the building track.

What we will learn today

1 Map the business process, and find the decisions in it

Draw any workflow as Inputs → Process → Decision → Action → Outcome — including the exception paths — then locate every point where a choice is made.

2 Judge whether AI is a good fit for each decision

Test each one: is it a written rule, ordinary software, or genuinely AI? Most are not AI, and saying so is the skill.

3 Design the AI application for the decisions that fit

What data it needs, which model fits, what number proves it worked, and where the human sits.

What you should be able to do by December

Two folds. The first is required of everyone. The second depends on the background you arrive with.

1 · Analyze — required of everyone

Take a real US industry use case — healthcare, finance, manufacturing, retail, transportation, government. Map the workflow, find the decisions, judge where AI genuinely fits, and defend that answer in a one-page proposal a manager could act on.

Fourteen of these become your course portfolio.

2 · Build — if your background supports it

If you arrive with some statistics, data, or programming experience, you will also build a small working prototype for one decision you analyzed — enough to demonstrate the idea, not a production system.

This is the stretch path on the final project, not a requirement for everyone.

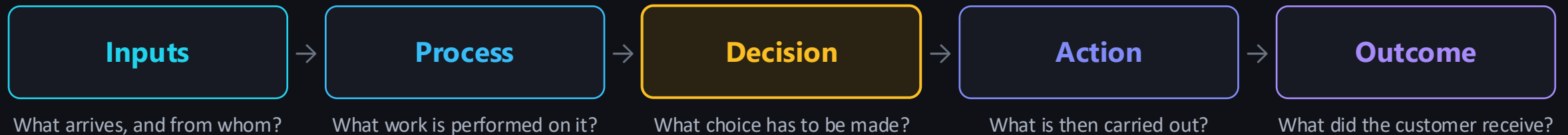
STEP 1 OF 3 · SECTION I

How work happens

You cannot judge work you cannot see. Before anything about AI, we reconstruct what actually takes place.

A business process converts an input into an outcome

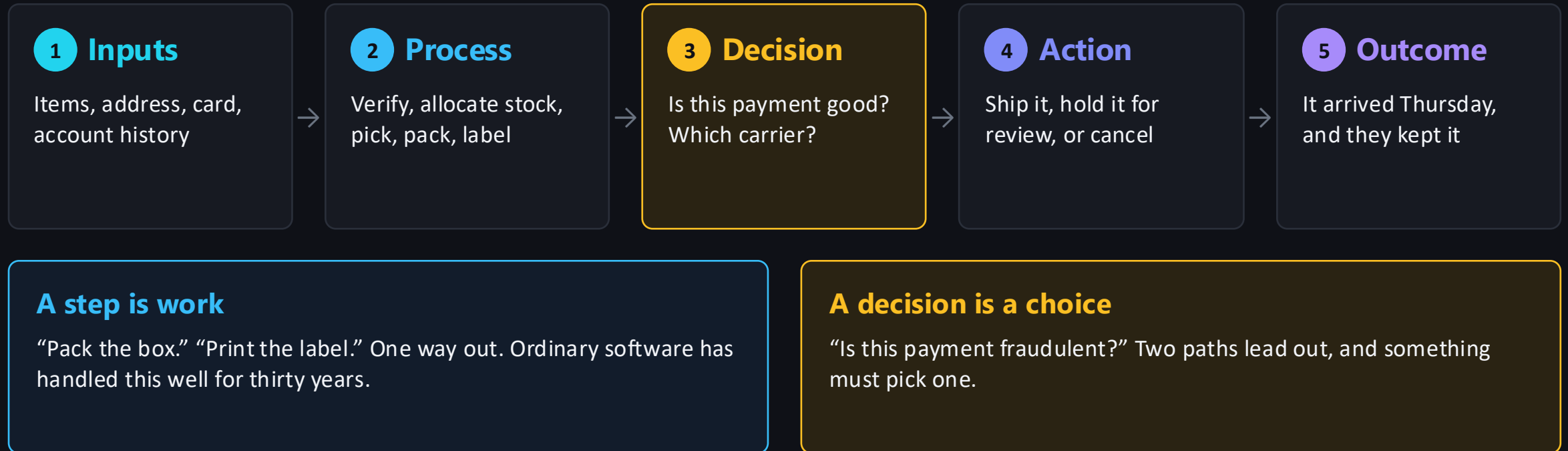
A repeatable sequence of steps that turns something arriving into something a customer values.



“Repeatable” is the load-bearing word

Forty thousand orders a month is a process. Doing something once is a project. You can only measure what happens again.

The same five stages, on our order — and only one is a choice

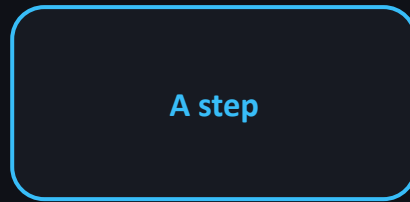


Let us review a workflow in our app

Five symbols, and what they are actually called



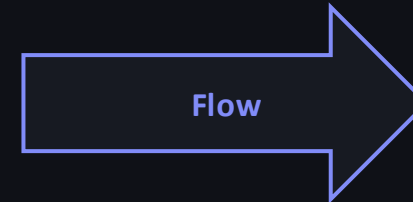
Terminator



Process



Decision



Flow line



Stored data

The diamond is the one that matters

The other four describe how work moves. The diamond is where a choice happens.

Label every arrow leaving a diamond

“Yes” and “no.” An unlabelled branch is the commonest mistake on a first map.

How to draw one properly — two references

ASQ · “What is a Flowchart?” — a seven-step procedure and the standard symbols asq.org/quality-resources/flowchart

Lucidchart · symbol reference with worked examples lucidchart.com/pages/flowchart-symbols-meaning-explained

Both are free. The formal standard, if anyone asks, is ISO 5807 — most people just say “workflow diagram.”

STEP 1 OF 3 · SECTION II

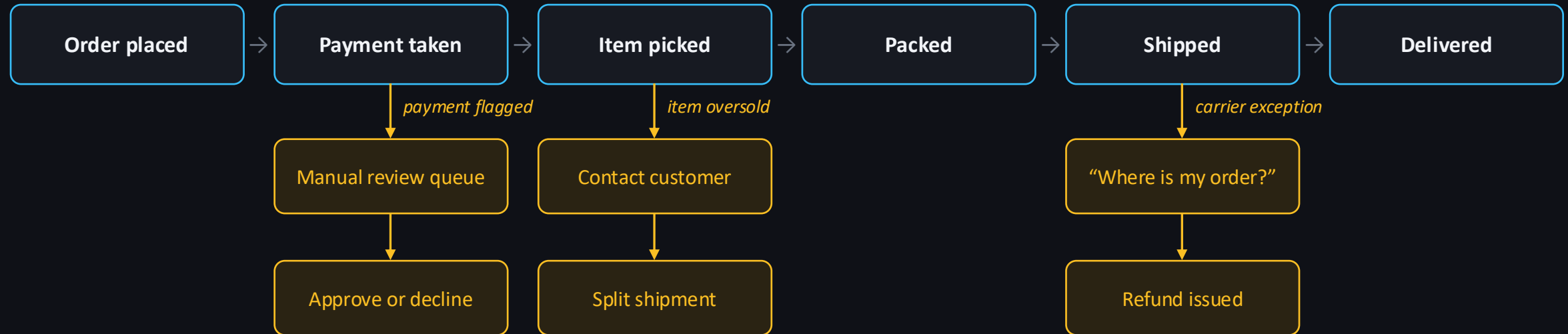
Mapping our use case

One online order, from Buy Now to the doorstep — first as it is documented, then as it actually runs.

The documented path, and what actually happens

 the documented path — what the written procedure says

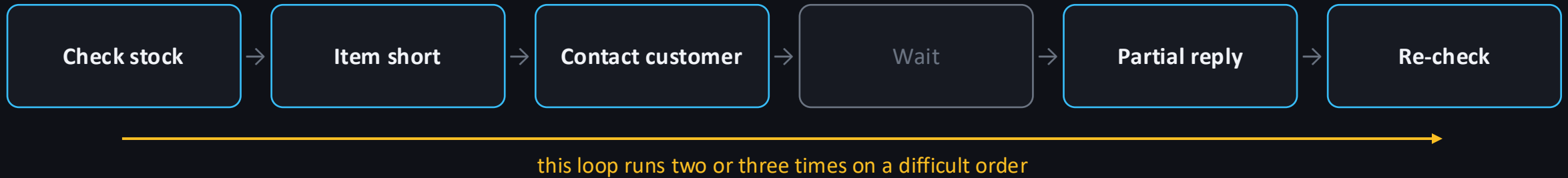
 what actually happens to roughly 1 order in 8



The top line is accurate and consumes almost no labour. Every branch below it does.

Proportions are a teaching assumption for this scenario, not an industry benchmark.

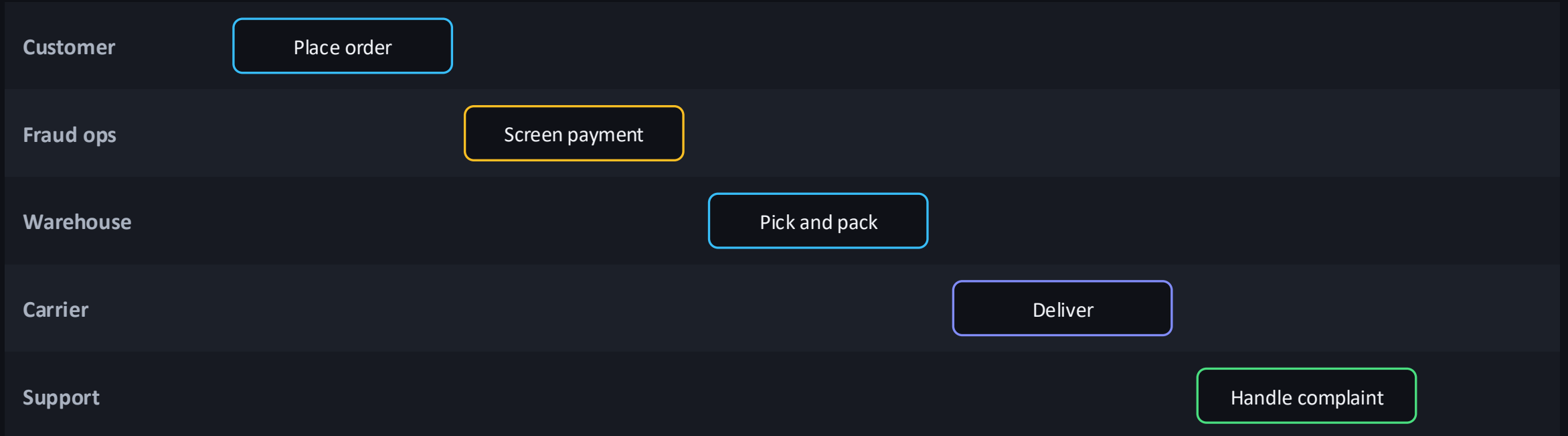
Follow one order through the rework loop



A map becomes analytical the moment it shows waiting and repetition

The boxes cost minutes. The loop costs days. The delay is in the waiting and the repeating, not in the checking.

A handoff is a queue, a contract, and a chance to lose context



Every lane crossing is a queue, and a point at which context is lost.

Add role bands to a flowchart and the result has its own name: a cross-functional flowchart, or swimlane diagram.

Six decisions inside a single order

THE DECISION	WHO DECIDES TODAY	HOW OFTEN	COST OF GETTING IT WRONG
Is this address deliverable?	System	Every order	A parcel goes nowhere
Is this payment fraudulent?	Fraud analyst	1,240 / month	A chargeback, or a good customer lost
Will it arrive when promised?	System	Every order	A broken promise and a support ticket
Which warehouse, which carrier?	System	Every order	Extra freight cost, or a late delivery
What do we tell an unhappy customer?	Support agent	~900 / month	Churn, or a refund we never owed
Refund, replace, or deny?	Support lead	~400 / month	Margin, or a customer who never returns

Volumes are teaching assumptions for this scenario, not industry benchmarks.

Quantify the baseline before you propose anything

40,000

orders per month

classroom assumption

3.1%

flagged for manual review

classroom assumption

4 min

per manual review

classroom assumption

1.8%

delivered late

classroom assumption



83 staff-hours a month on manual payment review alone

$40,000 \times 3.1\% \times 4 \text{ min} \div 60$

STEP 2 OF 3 · SECTION III

Where AI fits, and where it does not

Find every decision in the flow, then test each one — because most will not be AI problems, and saying so is the job.

Three different things, and the one question that separates them

Ordinary software

Stores, moves, and displays information. It decides nothing at all. Your order database is this.

A written rule

Executes a rule a person wrote down. “If the total is over \$2,000, route it to a lead.” Forever.

AI

Infers a rule nobody wrote down, from the pattern across thousands of past examples.

The test — would it behave differently with another year of data?

No means a written rule: a person fixed the logic in advance, so more history changes nothing. Yes means AI: the logic came out of the data, so different data produces different behaviour.

This is also why AI needs looking after and a written rule does not

A rule written in January still means what you meant in December. A model has quietly drifted, and nothing announces it.

Now run the test on all six decisions

THE DECISION	WHAT IT REALLY IS	WHY
Is this address deliverable?	A rule	The postal format is published. Write the rule.
Is this payment fraudulent?	AI · classification	Real logic that nobody has fully written down.
Will it arrive when promised?	AI · prediction	Depends on weather, volume, and this carrier's history.
Which warehouse, which carrier?	Optimization	Solvable exactly. Usually not machine learning at all.
What do we tell the customer?	AI, with a human	Drafting it is easy. Being right is the hard part.
Refund, replace, or deny?	Keep the human	Judgment, and the costs of being wrong are lopsided.

AI competes with four cheaper things, and often loses

THE ALTERNATIVE	IT WINS WHEN	IN OUR ORDER PROCESS
A rule	The logic is stable, known, and somebody will sign it	The address format check
Redesign the process	The step should not exist in the first place	Stop overselling instead of predicting it
Better ordinary software	The real problem is handoffs or visibility	Show support the carrier status directly
Do nothing at all	The current failure costs less than the fix	A 1.8% late rate may simply be acceptable

Select the least expensive intervention that resolves the problem.

STEP 3 OF 3 · SECTION IV

Data → Model → Metric → Human

Two decisions made it through the test. Only now is it worth asking how you would actually build one.

For the decisions that survive: four questions

1 Data

Does it exist, and can I actually get it?

Who owns it, is it labelled, and how long until it is usable?



2 Model

Which pattern, and which model underneath?

Classification, prediction, optimization, or generation.



3 Metric

What number proves it worked?

One technical criterion, and one a budget owner would sign.



4 Human

Where is the person when it gets it wrong?

Who reviews it, how often, and do they have time to look?

Data — does it even exist?

LEVEL	THE SITUATION YOU ARE ACTUALLY IN	REALISTIC TIME TO USABLE DATA
1	You own it, it is labelled, and it refreshes	Weeks
2	It exists and is labelled, but another team owns it	Add months. This is politics, not engineering.
3	You own it, but nobody has labelled it	Price the labelling before you promise anything
4	It was never captured — paper, phone calls, habit	You need instrumentation before any model
5	It does not exist	You are not doing AI. You are collecting data.

Model — four patterns, and that is very nearly all of it

Classification

Which bucket does this belong in? Out comes one of a known set of labels.

Prediction

What is likely to happen next? Out comes a number, or an event with a probability.

Optimization

What is the best arrangement? Out comes an assignment, a route, or a schedule.

Generation

Produce something new that fits. Out comes text, an image, code, or audio.

Classification — which bucket does this go in?

In our order: look at this payment and put it in one of three buckets — approve it, send it to a human for review, or decline it outright.

THE SAME PATTERN, IN THREE OTHER INDUSTRIES

Is this chest X-ray normal, or does it need a radiologist today?

Which of eleven teams should this incoming email be routed to?

Is this part coming off the line good, or is it defective?

What it needs: thousands of past examples where somebody already labelled the right bucket — and labelled it correctly.

Prediction — what is likely to happen next?

In our order: given the warehouse, the carrier, the route, and the time of year, will this parcel actually arrive by Thursday as promised?

THE SAME PATTERN, IN THREE OTHER INDUSTRIES

Will this patient be readmitted within thirty days of discharge?

How many units of this item will we sell in the next quarter?

Which customers are about to cancel their subscription?

What it needs: history where you know how things actually turned out — not merely what was decided at the time.

Optimization — what is the best arrangement?

In our order: out of nine warehouses and four carriers, which combination gets this parcel there on time for the least money?

THE SAME PATTERN, IN THREE OTHER INDUSTRIES

Which nurses work which shifts, given skills, leave, and labour rules?

What route should each of forty delivery vans take this morning?

How should this quarter's advertising budget be split across five channels?

What it needs: an objective you can state precisely, and constraints you can state precisely. Both are harder than they sound.

And most optimization is not machine learning at all

Routing, scheduling, and allocation were solved by operations research decades ago. A solver is cheaper, needs no training data, is provably optimal, and can be explained line by line to a regulator. When a vendor calls this AI, they are describing the packaging.

Generation — produce something new that fits

In our order: the customer has written in asking where their parcel is. Draft a reply that answers them, in our tone, with the right tracking details in it.

THE SAME PATTERN, IN THREE OTHER INDUSTRIES

Summarize a two-hundred-page case file into one readable page.

Draft six versions of ad copy for the same product launch.

Write the first version of the tests for this piece of code.

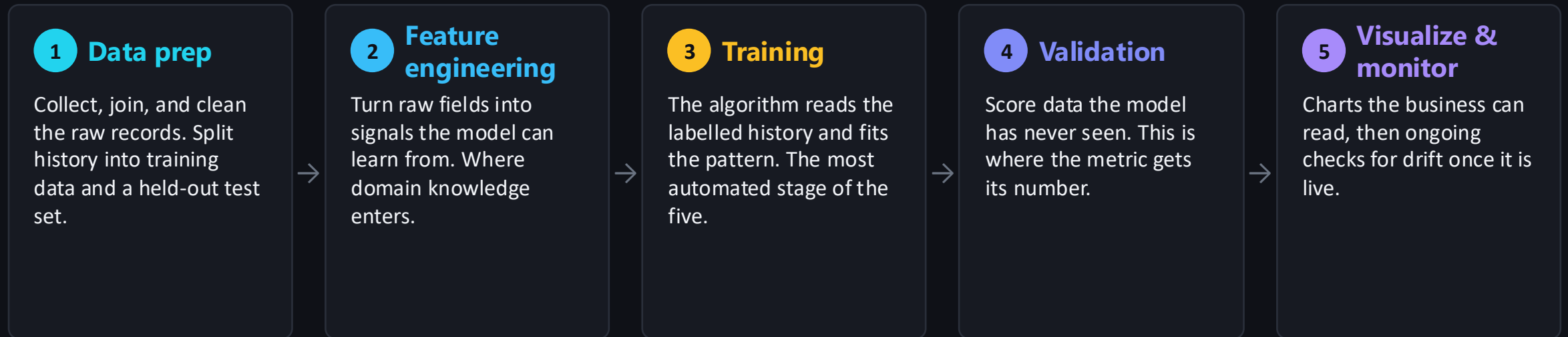
What it needs: something real to ground it in — actual policy, actual order data — and a human who checks the output before it goes out.

This is the one pattern that does the work rather than making the choice

The other three hand you a judgment that resolves a diamond. This one produces an artifact, inside the Process box. That is why it feels different — and why people reach for a chatbot when the problem was really classification.

How a model actually gets built

“Train a model” is itself a workflow. Data scientists follow the same five stages on nearly every project.



Where the time actually goes

Practitioners consistently report that most of a project is spent in the first two stages — preparing data and engineering features. Training, the stage everyone imagines, is often the shortest.

The same workflow, on our fraud decision

STAGE	ON THE PAYMENT-SCREENING MODEL	WHO IS INVOLVED
Data prep	Pull three years of orders. Join each to its outcome — chargeback, refund, clean. Hold out the most recent six months for testing.	Data engineer, with the fraud team
Features	Order amount vs. this customer's usual. Shipping and billing address mismatch. Account age. Device seen before. Time of day.	Data scientist + fraud analysts
Training	Gradient boosting fits the pattern across ~1.4 million labelled orders.	Data scientist
Validation	Score the held-out six months. Does it catch 85% of known fraud while flagging under 3% of good orders?	Data scientist, verified by the business
Visualize & monitor	A score-distribution chart the fraud team can read, a threshold they can move — and a monthly drift check.	Everyone. This is the handover.

Metric — technical success and business success are different

The technical criterion

Catch 85% of the fraud we catch now, sending under 3% of good orders to review. Measured on orders it has never seen.

The business criterion

Halve the 83 monthly review hours, without chargebacks rising and without turning away good customers.

A model can be excellent and still worthless

The second is the one a budget owner reads — and the one most often left unwritten.

Each pattern is measured by its own number

PATTERN	THE TECHNICAL METRIC	WHAT IT ACTUALLY MEANS	THE TRAP
Classification	Precision · Recall	Precision: of the payments we flagged, how many were really fraud. Recall: of the real fraud, how much did we catch.	Accuracy misleads when the thing is rare. Call every order legitimate and you are “99% accurate.”
Prediction	MAE · RMSE (MSE)	The average size of the error, in the unit you care about — days late, units sold. RMSE punishes large misses harder.	An average hides the expensive outliers. Look at the worst 5%, not the mean.
Optimization	Cost against the baseline	Freight spend, hours scheduled, distance driven — measured against what you do today, not against theory.	Optimal against the wrong objective is still wrong. Check who wrote the objective.
Generation	Groundedness · acceptance rate	Groundedness: is every claim traceable to a real source. Acceptance: how often the human sends the draft unedited.	It reads beautifully and is confidently wrong. Fluency is not accuracy.

Every one of these still needs a business number beside it. “Precision of 0.92” is not a result.

Human — where is the person when it gets it wrong?

A false positive

You blocked a real customer. They are annoyed, and may never come back.

A false negative

Fraud got through. You lose the goods, the payment, and a chargeback fee.

Which is worse?

A business decision, not a technical one. The domain expert answers it.

And can the reviewer actually tell?

Clearing 400 flagged orders an hour is clicking approve, not reviewing. The checkpoint exists on the diagram and nowhere else.

ALL THREE STEPS · SECTION V

One decision, all the way through

Both flows, applied to a single decision from the order we mapped tonight. This is the shape of every assignment you will hand in.

Payment screening, end to end

Problem

1,240 payments a month are flagged and reviewed by hand, at four minutes each — 83 staff-hours. Analysts apply judgment nobody has fully written down.

Solution

A classifier ranks each flagged payment. Clear cases are approved or declined automatically; genuinely ambiguous ones go to an analyst.

Data

Three years of orders with their outcomes — chargebacks, refunds, confirmed fraud. We own it, and it is labelled. Level 1.

Model

Classification into three buckets: approve, review, decline. Gradient boosting on tabular features. Explainable enough to show a regulator.

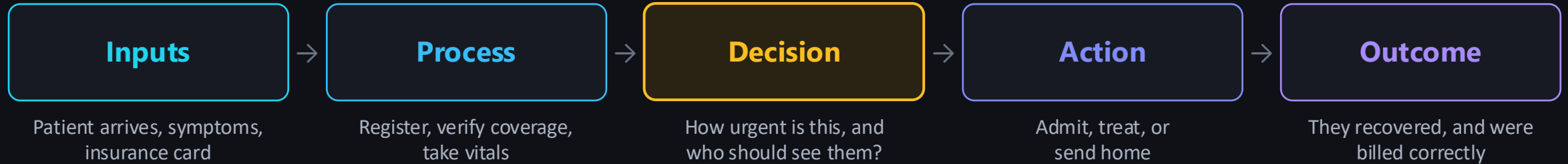
Benefits

Review hours cut from 83 to roughly 40. Faster decisions for good customers. Analyst attention redirected to the ambiguous cases.

Challenges

Labels record what analysts decided, not what was true. The model drifts as fraud patterns move. The reviewer must have time to genuinely review.

The same five stages, on a hospital intake



The questions transfer. The answers do not.

Same stages, same diamonds, same four patterns. But error costs are measured in patient harm, a regulator is in the room, and the human stays closer to the loop.

Key takeaways

1 Step 1 — hunt for the diamonds

Four of the five stages are work; only the third is a choice. Three of the four AI patterns land there — generation is the exception, and works inside a step.

2 Step 2 — most decisions are not AI problems, and saying so is the job

Four of our six were eliminated — by a rule, by a solver, or by keeping the human. The eliminations were the finding.

3 Step 3 — four patterns, and four questions

Classification, prediction, optimization, generation. Then Data, Model, Metric, Human.

4 Underneath all three

Select the least expensive intervention that resolves the problem. A model where a rule would do is not a solution.